

Jank: This term is often used in the mobile apps space, and refers to the effect of slow or inefficient code operations in an app, which block the updating of the UI and cause it to lag or become unresponsive.

Firefox OS build prerequisites

Here is a small guide on how to build the Boot 2 Gecko project's source code.

Requirements for GNU/Linux

To build the B2G project source code on Linux, you need the following minimum requirements:

- A 64-bit GNU/Linux distribution
- At least 4GB RAM
- Free hard disk space of 30GB

This is more than the bare minimum requirements, but sometimes the build might fail just because of some missing resources. A typical error in this case is 'arm-linux-androideabi-g++: Internal error: Killed (program cc1plus)'.

You will also need the following tools installed in the system before you start the build process:

```
autoconf 2.13, bison, bzip2, ccache, curl, flex, gawk, git,
gcc / g++ / g++-multilib, java sdk (jdk), make, OpenGL
shared libraries, patch, X11 headers, 32-bit ncurses, 32-bit
zlib
```

Requirements for 64-bit installation

Ubuntu/Linux Mint: Run the following commands in a terminal:

```
sudo apt-get update
sudo apt-get install autoconf2.13 bison bzip2 ccache curl
flex gawk gcc g++ g++-multilib git ia32-libs lib32ncurses5-
dev lib32z1-dev libgl1-mesa-dev libx11-dev make zip
```

If you are building for the Flame reference phone or for Nexus 5, run the following command in a terminal:

```
Sudo apt-get install libxml2-utils
```

In addition to the emulator build issues discussed above, the compiler will default to `gcc-4.7`, which will fail to build due to an error along the following lines:

```
"KeyedVector.h:193:31: error: indexOfKey was not declared
in this scope, and no declarations were found by argument-
dependent lookup at the point of instantiation"
```

In a fresh Ubuntu 12.10 install, you'll get an error about unmet dependencies for `ia32-libs`. The following commands fix it:

```
sudo dpkg --add-architecture i386
```

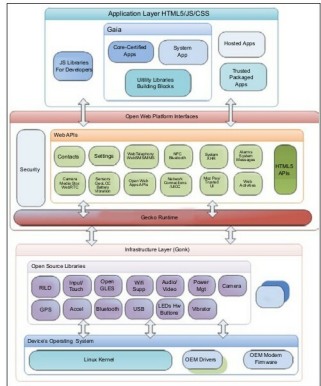


Figure 1: Architecture

```
sudo apt-get update
sudo apt-get install ia32-libs
```

Fedora 19/20: Those using Fedora 19/20 can type the following commands in a terminal:

```
sudo yum install autoconf213 bison bzip2 ccache curl flex
gawk gcc-c++ git glibc-devel glibc-static libstdc++-static
libX11-devel make mesa-libGL-devel ncurses-devel patch zlib-
devel ncurses-devel.i686 readline-devel.i686 zlib-devel.i686
libX11-devel.i686 mesa-libGL-devel.i686 glibc-devel.i686
libstdc++-i686 libXrandr.i686 zip perl-Digest-SHA wget
```

The project should build correctly using the default compiler shipped with Fedora but if you encounter compilation errors, you might need to install `GCC 4.6.x`:

Download the *tarball* and then install it to `/opt` with the following command on Fedora 19/20:

```
curl url -O http://people.mozilla.org/~gsvelto/gcc-4.6.4-
fc19.tar.xz
sudo tar -x -a -C /opt -f gcc-4.6.4-fc19.tar.xz
```

Arch Linux: To install the required resources in Arch Linux, run the following command in the terminal:

```
sudo pacman -S --needed alsa-lib autoconf2.13 bison ccache
curl firefox flex gcc-multilib git gperf libnotify libxt
```